

2026 S2 ENGG3000

SPINE Engineering Project

Electrical and Software Engineering

“Whack a Mole” Game

Version 2.1

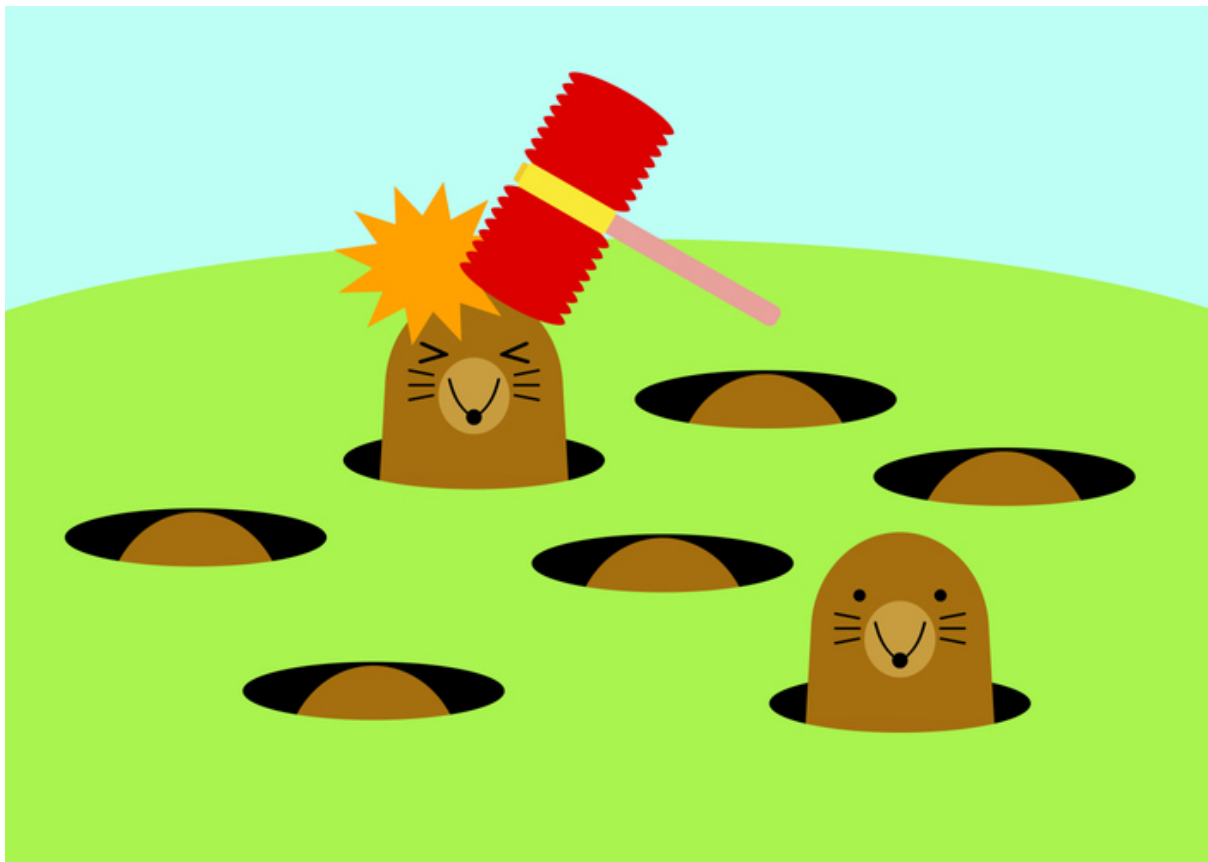


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1. Document History

Revision	Release Date	Author	Changes
1.0	27/07/2026		Initial Release
2.0	05/08/2026		Changes indicated in Yellow text, as follows: • Updated area size for the physical space. • Sensor details/requirements updated. • Budget details added. • Power supply details updated.
2.1	17/08/2026		Changes indicated in Green text, as follows: • Minor change to the size of the boxes (to accommodate the extra batteries that were added in Version 2.0) • Minor change to the pricing of the sensors (to match the advertised price from the suppliers)

2. Project Details

The Group project is to design and build a full body “Whack a Mole” game. Based on the classic Japanese arcade game (<https://en.wikipedia.org/wiki/Whac-A-Mole>), your game will be played with the “moles” randomly appearing on a video screen. The player positions a cursor on the screen by moving their whole body in front of the screen, with the system tracking the position of the player’s body **using only ultrasound sensors**. If the player moves their body over the “mole” before it disappears, they score a point, otherwise a new mole pops up somewhere new.

Each group must **design, prototype, build** and **iterate** their product using Engineering Principles.

The system must be able to sense and track a person over an open Playing Area of **1.5m wide x 1.4m deep, in front of a ‘dead zone’ that is in front of a screen that is on a wall. The dead-zone extends 60cm out from the wall. All components of the system must be located within 50cm of the wall.** No part of the system can be within the Playing Area. The system should emit an audible alarm if the person **enters the dead-zone** to prevent players running into the screen. The alarm should be accompanied by an effective visual warning.

The system can consist of multiple sensors. Each sensor must be a standalone box no larger than 100 mm x 100 mm x **150 mm**. They must run for 1 hour using **4 x AA Nickel Metal Hydride batteries** as their only power source. Each box must communicate wirelessly with the PC running the game. **Ideally, the boxes should be rechargeable from a standard USB-A port (but this is not essential).**

You can be creative in your development of the game, but it must remain in the spirit of the original Whack a Mole game. The game should have multiple levels of increasing difficulty and be delivered by a software package that can be downloaded and installed on a Windows laptop.

3. Product Demonstration

In Week 5 and Week 9, your team should demonstrate progress towards full functionality with a particular emphasis on integration of sub-systems towards meeting the overall function of the system. By Week 13, your team should also be able to demonstrate the performance benchmarks and meet the design and manufacture requirements. The assessment of the Whack-a-Mole system will be based on the following criteria:

1. Full Functionality

- Sends sensor data wirelessly to the PC
- Runs on supplied batteries
- Integrates sensor data into calibrated spatial positioning
- Alarm when within 60cm of screen
- Original Whack-a-Mole game with levels
- Play the Whack-a-Mole game based on sensor data

2. Performance Benchmarks

- Sensors less than 100 mm x 100 mm x 150 mm
- Current consumption and battery data show life > 1 hour
- Person tracking is sufficiently accurate to play game
- Person tracking is sufficiently responsive to play game

3. Design and Manufacture

- Code can be downloaded and installed on a Windows laptop.
- Sensors have neat external appearance
- Sensors have sound electronic construction
- Supplied components are returned in good working order
- No solder, adhesives, paint or other like material has been applied to the supplied components

4. Components/Materials/Approved Suppliers

The following components can be used (and will be supplied):

- 2 x ESP32 processor board (\$8 each)
- 2 x Antennas (\$0 each)
- 4 x Ultrasonic sensor – RCWL-1601 (\$2.70 each)
- 2 x Battery Pack (each pack incl. 4 AA NiMH batteries and a Battery holder) (\$7 each - discounted price)
- Perf board
- Connectors

Additional numbers of each of these components are available from the store. They will be included in your budget at the above prices, except for the batteries which will be charged at \$6.50 each, and the battery holders \$3 each.

Other components can be sourced from the approved suppliers listed below. There will be one opportunity during the semester to submit a list of desired components, that will be ordered for you. When considering your order, you should take note of stock availability and lead times for restocking. It is highly recommended that you only request components that are in stock and have a reasonable quantity remaining.

The overall budget for your project (including the components that have been supplied to you) is \$100.

Supplier	URL
Bunnings	https://www.bunnings.com.au/
RS Online	https://au.rs-online.com/web/
Core Electronics	https://core-electronics.com.au/
Jaycar	https://www.jaycar.com.au/

5. Assessment Tasks

Assessment 1: Sprint 1 – Minimum Viable Product (MVP)

Timing: End of Sprint 1 (Week 5)

Weight: 25% Total (5% Group, 20% Individual)

Prototype Generation: Prototype V1 (Proof-of-Concept)

Development of an early, low-fidelity, fully integrated system that addresses the absolute core value proposition of the customer brief. It doesn't need to be optimized or robust, but it must complete the primary, baseline loop of the system in front of the client.

Deliverables

- **Prototype V1 Live Demonstration (Group 5%):** A functional proof-of-concept proving the core loop works. The team must explicitly show how this low-fi hardware baseline directly solves the customer's immediate problem.
- **Peer Assessment (Individual 5%):** Individual contributions to the demonstration will be recognised through a transparent, normalised peer evaluation process.
- **Individual Traceability & Ideation Log (Individual 15%):** Documentation of the creative thinking strategies used to move from raw requirements to physical Prototype V1 hardware, tracking how the student's assigned subsystem directly contributed to the MVP's customer value.

Assessment 2: Sprint 2 – Critical Design Review

Timing: End of Sprint 2 (Week 9)

Weight: 35% Total (10% Group, 25% Individual)

Prototype Generation: Prototype V2 (Ruggedized & Subsystem-Optimized)

Development that builds on the Prototype V1 and systematically optimizes its parameters. Prototype V2 must demonstrate a step-change in technical quality: ruggedized subsystems, tighter tolerances, reliability and the integration of secondary customer features.

Deliverables

- **Prototype V2 Live Demonstration (Group 10%):** The team must demonstrate Prototype V2, presenting empirical data that shows how much its performance metrics have improved compared to the V1 baseline.

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- **Peer Assessment (Individual 10%):** Individual contributions to the demonstration will be recognised through a transparent, normalised peer evaluation process.
- **The Engineering Rigour & Change Log (Individual 15%):** A technical document detailing the mathematical modelling, simulation calibration, and physical testing of the team member's specific subassembly. It must include a formal Change Register proving how lessons learned from V1's failures were tracked and resolved in V2. An honest evaluation of personal knowledge gaps discovered during the V1-to-V2 transition.

Assessment 3: Sprint 3 – The Final Product

Timing: End of Sprint 3 (Week 13)

Weight: 40% Total (15% Group, 25% Individual)

Prototype Generation: Prototype V3 (Final Asset)

Development of the final product. Prototype V3 must be completely integrated, enclosed, and validated under real-world stress conditions. The customer value must now be defended to non-technical stakeholders, proving that the technical choices align with best engineering practice.

Deliverables

- **The Performance Stress-Test (Group 15%):** Live technical validation where Prototype V3 is run under strict operating conditions. Marks are awarded strictly on how effectively the integrated product hits the objective target parameters specified in the Product Description.
- **Peer Assessment (Individual 15%):** Individual contributions to the demonstration will be recognised through a transparent, normalised peer evaluation process.
- **The Public Exhibition Pitch (Individual Oral 10%):** Across 5 x 3 minute presentations, the team pitches Prototype V3 to a dual audience:
 - *The Technical Experts:* Defending the structural, electrical, and data integrity of the final product.
 - *The Customers/Lay-People:* Using simple, narrative-driven language to explain how the product meets user needs and its advantages over alternative solutions.